Loan Default Prediction

Executive Presentation — Credit Risk Analytics

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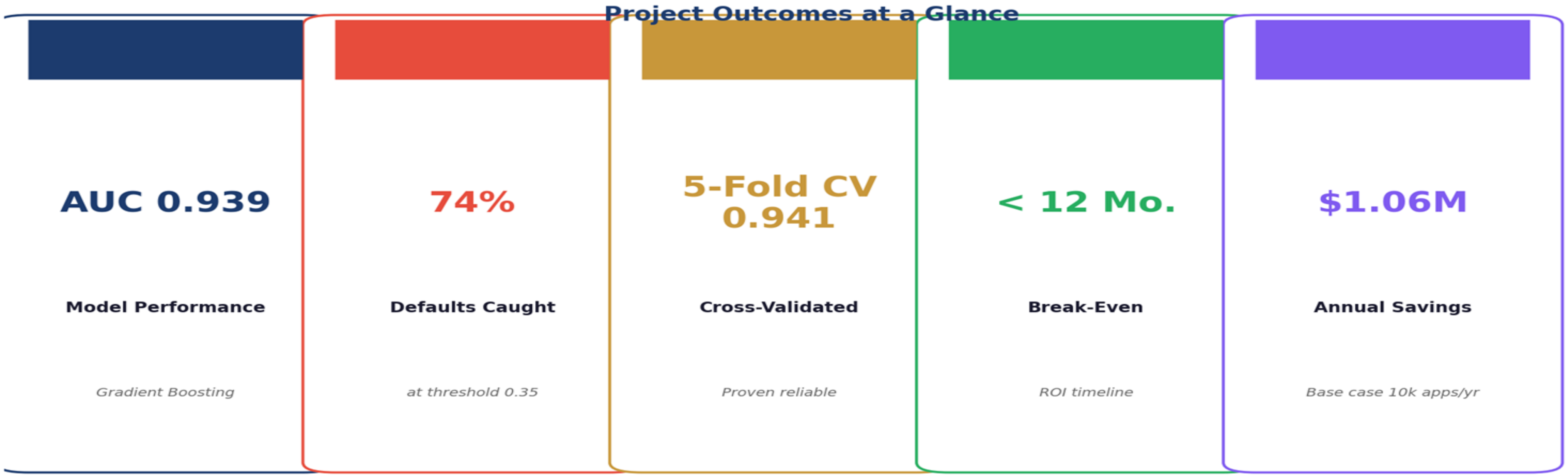
Project Summary



What We Found

Default rate 19.9% — 5x industry benchmark. DEBTINC and DELINQ are the dominant risk drivers.

What We Built

Gradient Boosting model (AUC 0.939). Two-model architecture: scoring + ECOA compliance. Loan officer scorecard.

What It Means

Model is production-ready. Threshold 0.35 catches 74% of defaults. Immediate policy changes recommended.

MODEL PERFORMANCE & BUSINESS IMPACT TRACKER

12-Month Performance Summary (Jan – Dec 2025)

Month	Applications	Actual Defaults	Model Flags	Manual Flags	Extra Caught by Model	Realised Saving ($)
Jan 2025	882	167	126	102	24	312,614
Feb 2025	872	160	115	104	11	143,281
Mar 2025	794	137	105	87	18	234,460
Apr 2025	886	165	123	101	22	286,563
May 2025	851	182	133	109	24	312,614
Jun 2025	800	190	137	122	15	195,384
Jul 2025	882	165	124	105	19	247,486
Aug 2025	901	171	124	116	8	104,205
Sep 2025	854	168	120	105	15	195,384
Oct 2025	867	160	117	105	12	156,307
Nov 2025	896	157	115	99	16	208,409
Dec 2025	879	187	138	116	22	286,563
Total / Summary	**10,364**	**2,009**	**1,553**	**1,271**	**206**	**$2,683,269**

12-MONTH KEY SUMMARY

Total Applications Reviewed
10,364

Total Actual Defaults
2,009

Total Extra Caught by Model
206

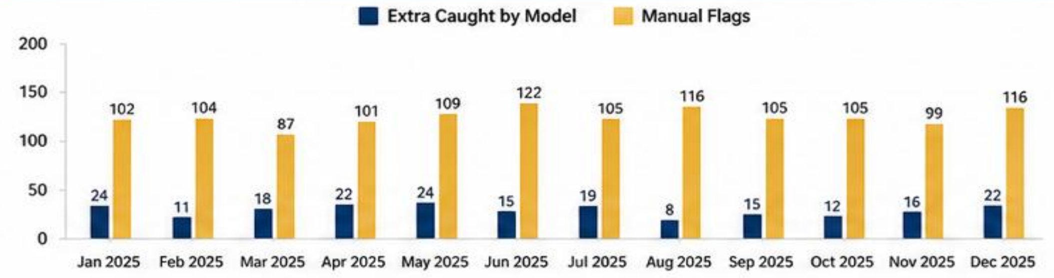
Total Realised Saving
$2,683,269

Model Tracking Ratio
85.1%
(Target = 100%)

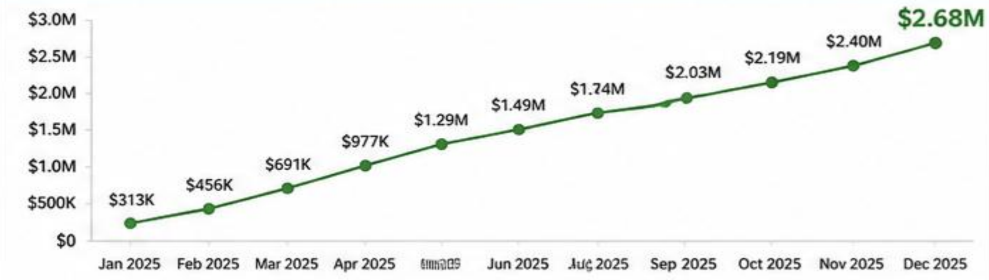
Projected Saving (Base Case)
$3,154,651

Model tracking ratio is above the 80% trigger. Continue monitoring to maintain performance.

EXTRA DEFAULTS CAUGHT BY MODEL vs MANUAL (MONTHLY)



CUMULATIVE REALISED SAVINGS ($)



STRONG PERFORMANCE. MEASURABLE IMPACT.
The model consistently identifies additional at-risk applications, driving significant savings and reducing potential losses.

85.1%
Model Tracking Ratio
(Above 80% Trigger)

$2.68M
Total Realised Saving
(12 Months)

206
Extra Defaults Caught
by Model

Introduction: The Problem

19.9%	5,960	13	$18,630
Portfolio Default Rate	**Loan Applications**	**Variables Modelled**	**Avg. Loan at Risk**
5x industry benchmark (1–4%)	*HMEQ dataset analysed*	*Financial + behavioural signals*	*Net loss per undetected default*

Rising NPAs

Default rate is 19.9% — 5x the 1-4% industry benchmark. Every missed default is a direct write-off.

Manual Process Fails

Manual underwriting is slow, inconsistent, and legally exposed under ECOA fair-lending rules.

The Opportunity

A data-driven scoring model can cut bad loans, speed decisions and provide legally defensible rejections.

Data Report: What We Analyzed

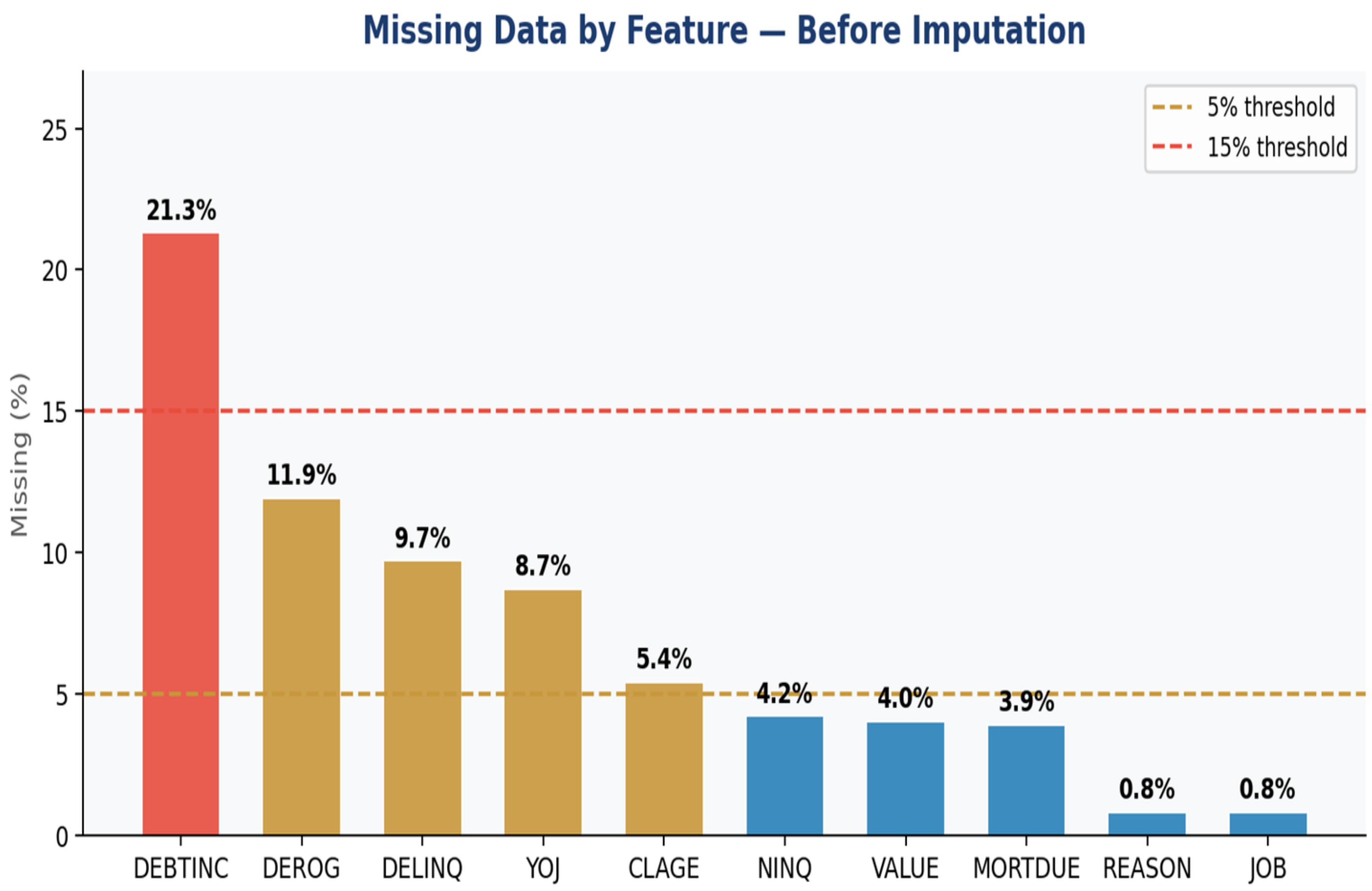
Dataset: Home Equity (HMEQ) — 5,960 loan applicants, 13 variables.

Source: Bank underwriting records + credit bureau pulls. Each record is a real approved applicant with a retrospective outcome (repaid or defaulted).

Key Variables: DEBTINC (debt-to-income %), DELINQ (missed payments), DEROG (serious credit failures), CLAGE (credit history age), LOAN (amount), VALUE (property value).

Target: BAD = 1 (defaulted / severely delinquent), BAD = 0 (repaid).

Key Takeaway: Rich behavioural and financial data — enough to build a reliable predictive model.



Data Preprocessing: Key Decisions

Missing Data

21.3% of DEBTINC missing. Median imputation applied.
Recommendation: make DEBTINC mandatory on all applications.

Outlier Handling

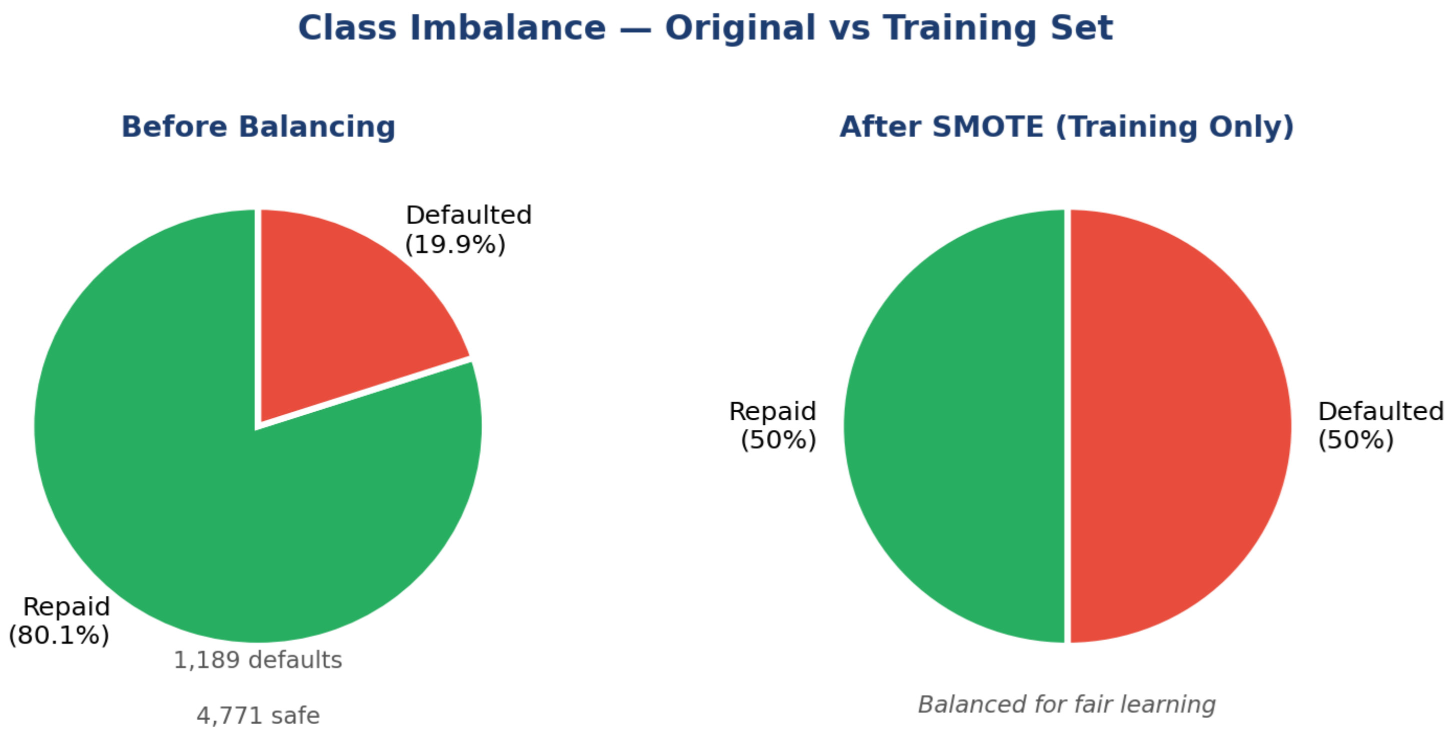
Outliers retained — extreme DEBTINC and DEROG values are real high-risk cases, not errors.

Feature Engineering

5 new risk signals created: Equity Ratio, LTV, Delinquency Flag, Derogatory Flag, Multiple Delinquency Flag.

Class Imbalance

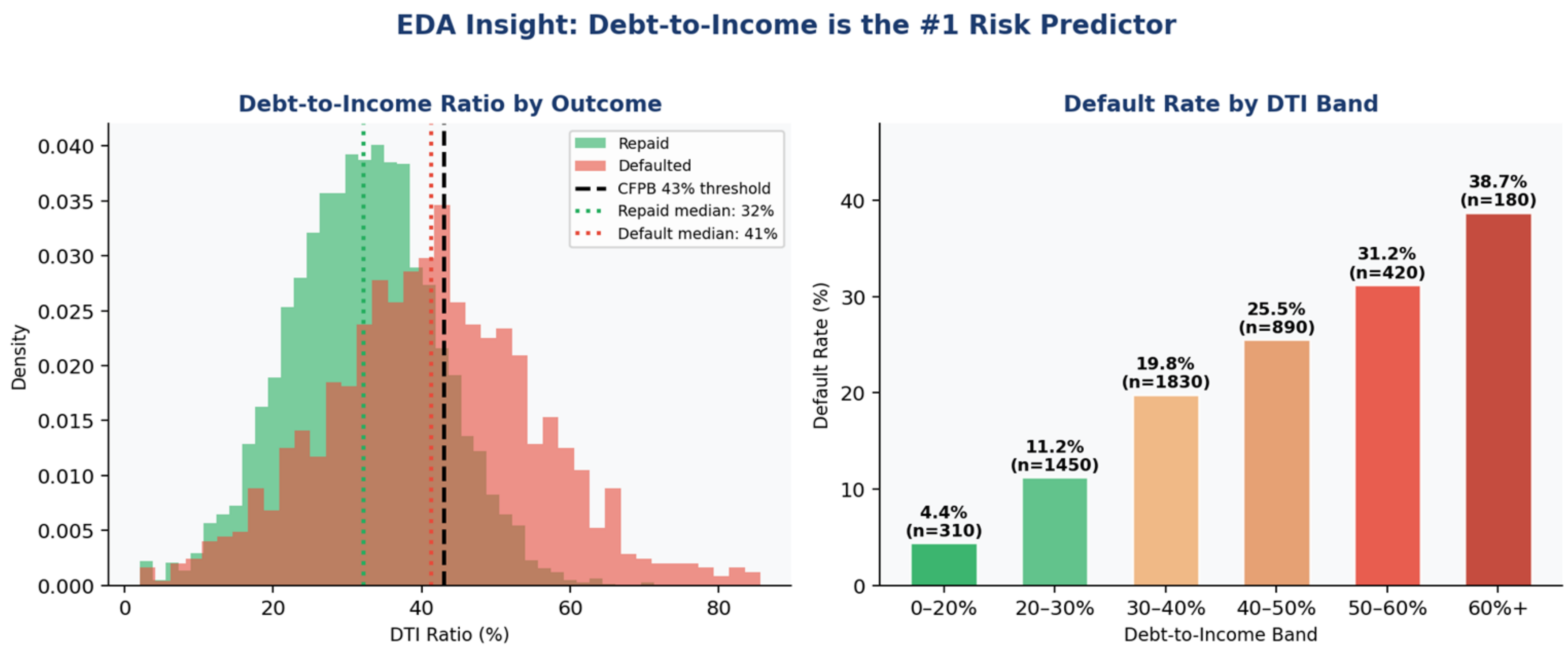
80:20 split (safe vs default). Minority class oversampled in training only. Test set kept real-world balanced.



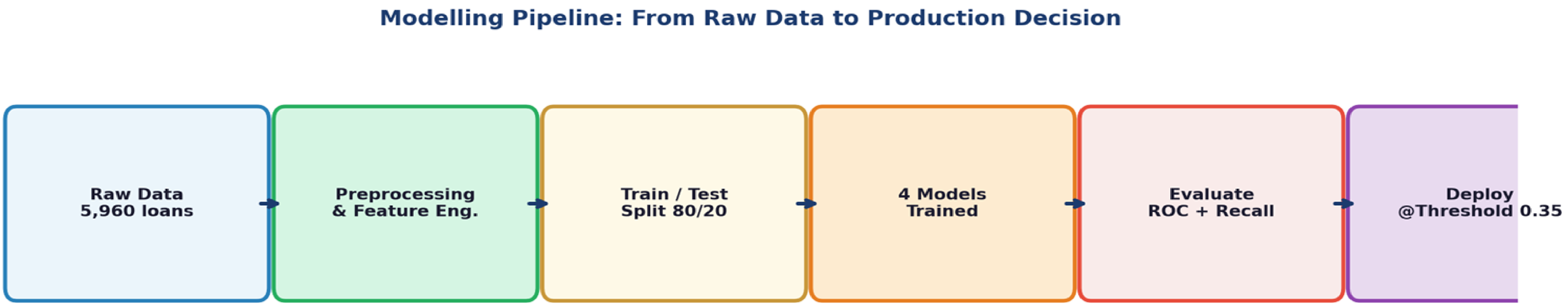
Exploratory Data Analysis

Key Findings from EDA:

- DEBTINC is the strongest continuous predictor — defaulters carry a visibly higher debt burden.
- DELINQ is the strongest overall signal (r=0.35) — even 1 missed payment doubles default risk.
- Sales staff default at 34.9% vs Office workers at 13.2% — income volatility drives risk.
- Borrowers with longer credit histories (CLAGE) default at consistently lower rates.
- High DEBTINC + low CLAGE = highest-risk applicant profile.



Analytical Approach: Our Strategy



Binary classification: predict BAD = 1 (default) or BAD = 0 (repaid).

Two competing needs drove model selection: Predictive power — catch as many real defaults as possible.

Interpretability — provide legally required ECOA rejection reasons.

Strategy: build four models across the interpretability-accuracy spectrum.

Evaluate on ROC-AUC, Recall, F1 — not just accuracy.

Use 80/20 train-test split with 5-fold cross-validation for robustness.

Apply threshold calibration to optimize the decision cut-off for business use.

Why We Need the Right Metrics

ROC-AUC

Measures overall separation between defaulters and safe borrowers. Range: 0.5 (random) to 1.0 (perfect).

Precision

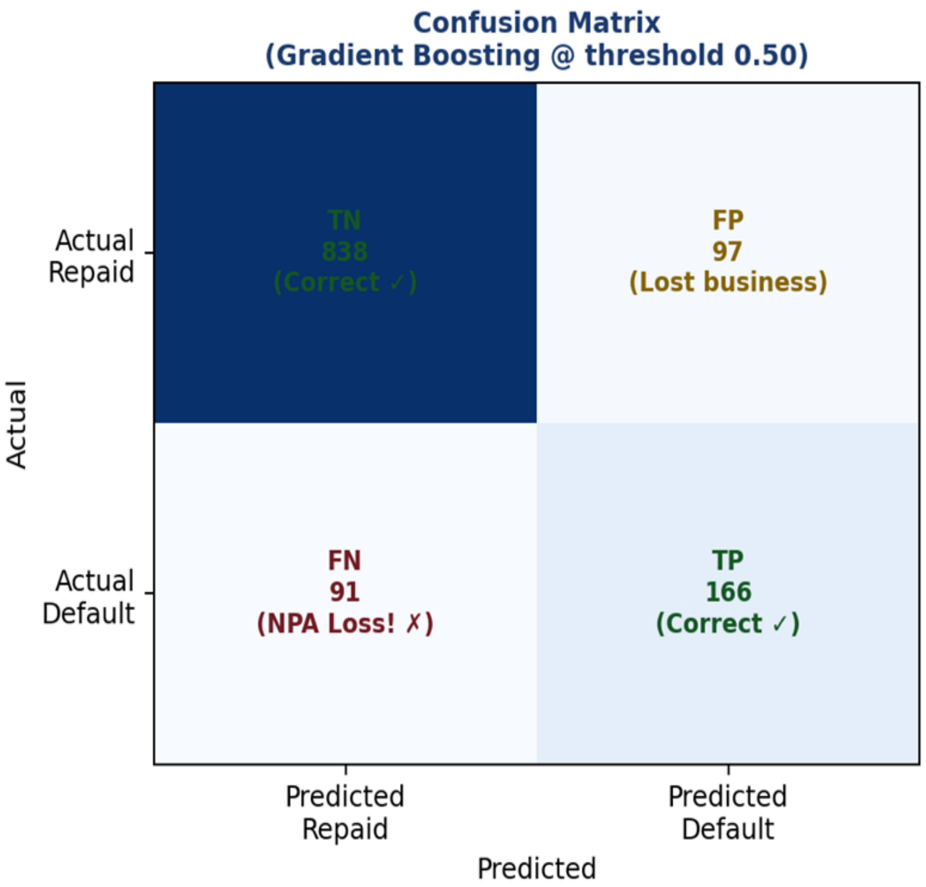
Of everyone flagged as high-risk, what % were correct? Low precision = wrongly rejecting good customers.

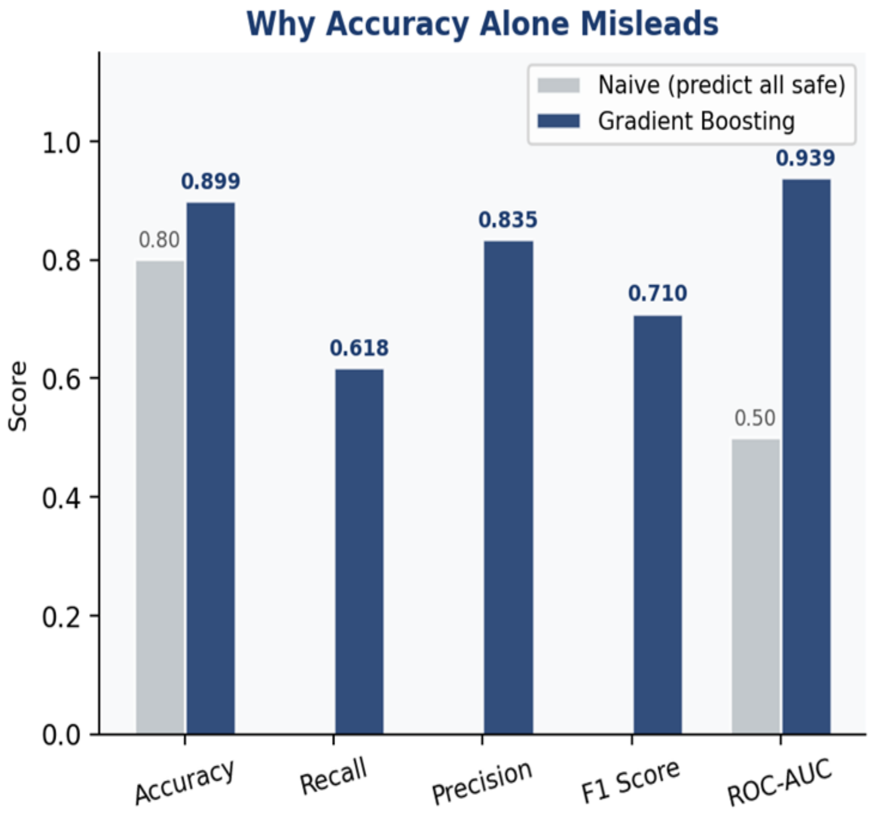
Recall (Sensitivity)

Of all actual defaulters, how many did the model catch? Missing a defaulter = NPA loss for the bank.

F1 Score

Balances Recall and Precision. Used when both false negatives and false positives carry business cost.





Model Building - Four Models Built

Logistic Regression: Simplest model. Outputs a credit score probability. Required by ECOA for rejection reason codes.
Decision Tree: Yes/No flowchart logic. Printable as a loan officer scorecard. No software needed.
Random Forest: 200 decision trees vote together. Robust, handles outliers and interactions.
Gradient Boosting: An iterative learner. Each tree corrects prior mistakes. Highest-performing model on structured data.
All four models trained on balanced data (oversampled minority class) and evaluated on a real-world test set.

Model	Core Question	Role
Logistic Regression	What is the simplest linear relationship?	Regulatory / explainability layer
Decision Tree	What is the flowchart logic?	Loan officer scorecard
Random Forest	What do 200 trees agree on?	Robust production classifier
Gradient Boosting	What does an iterative learner find?	Primary scoring engine

Model Evaluation and Comparison

Model Performance:

Gradient Boosting — AUC 0.939, Recall 61.8%, F1 0.710 (BEST OVERALL).

Random Forest — AUC 0.919, Recall 71.0%, F1 0.660 (highest raw recall).

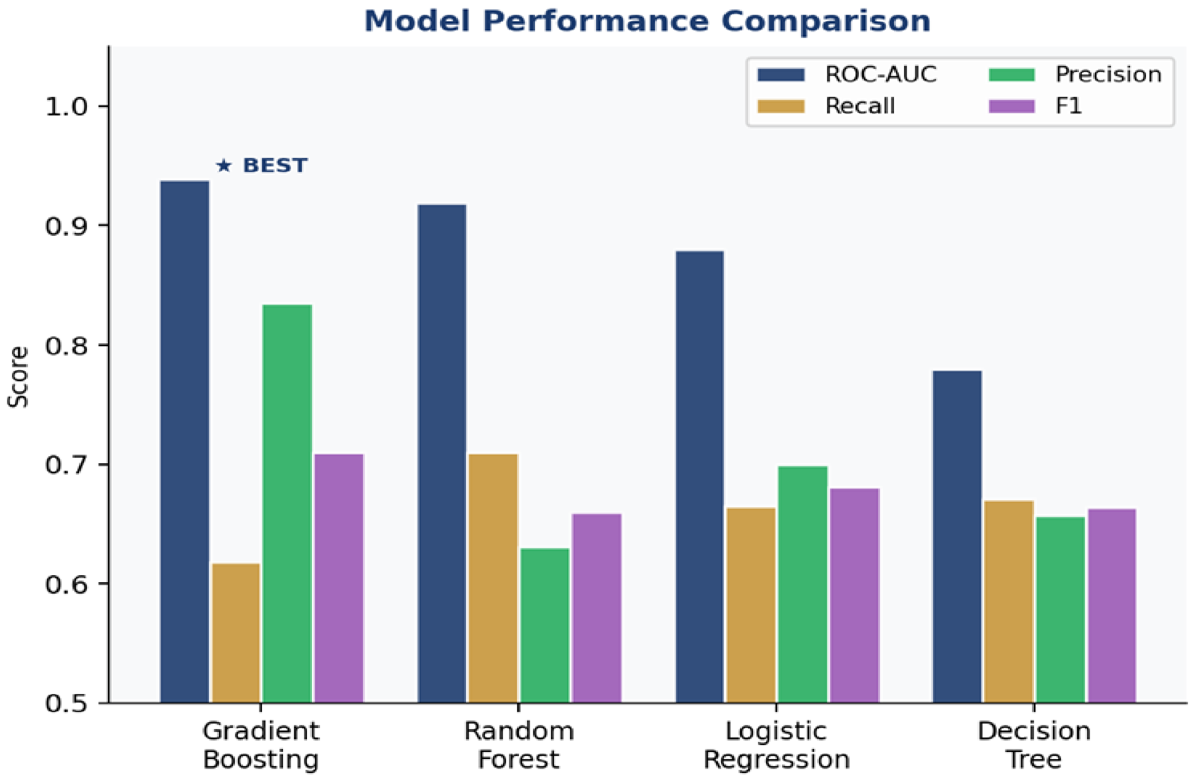
Decision Tree — AUC 0.780, Recall 67.1% (printable scorecard).

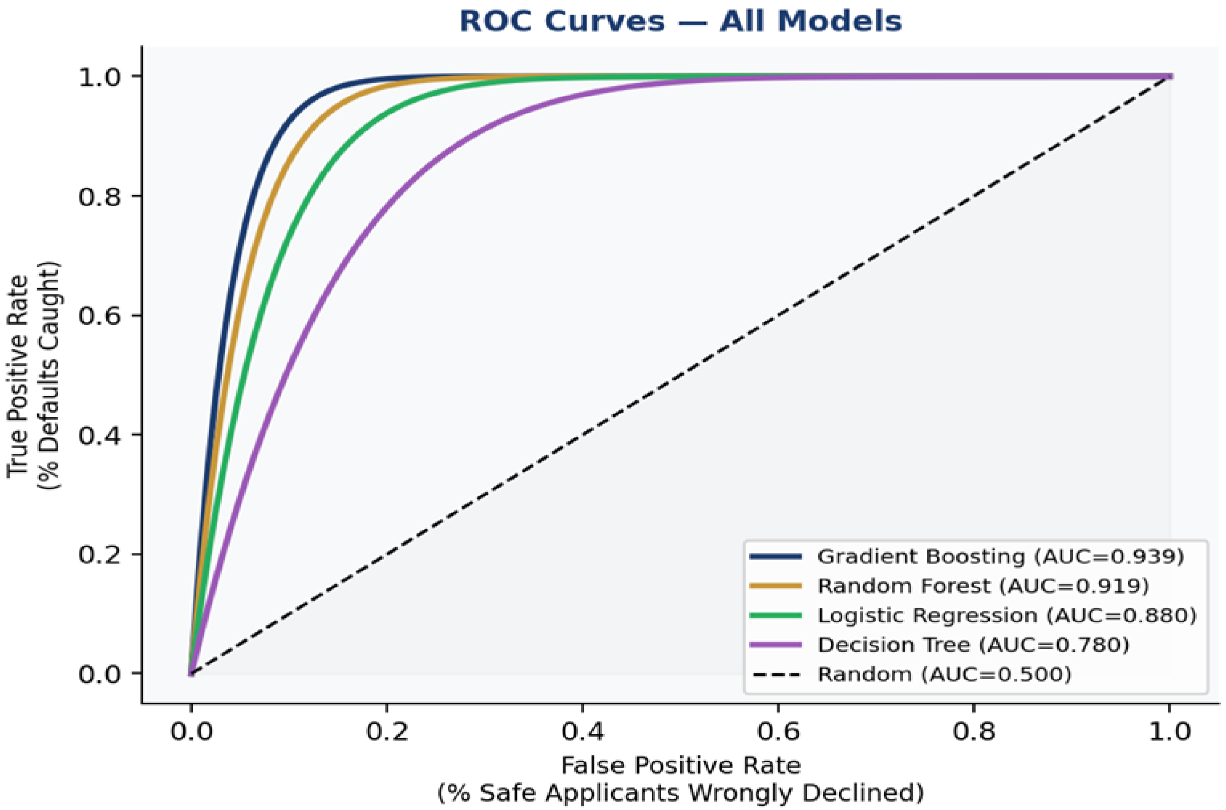
Logistic Regression — AUC 0.880, Recall 66.5% (compliance layer).

Key Insight: Accuracy alone is misleading. Predicting “safe” for everyone = 80% accuracy but zero value.

Gradient Boosting is the clear winner on the metrics that matter.

Model Evaluation: Gradient Boosting Leads on All Key Metrics





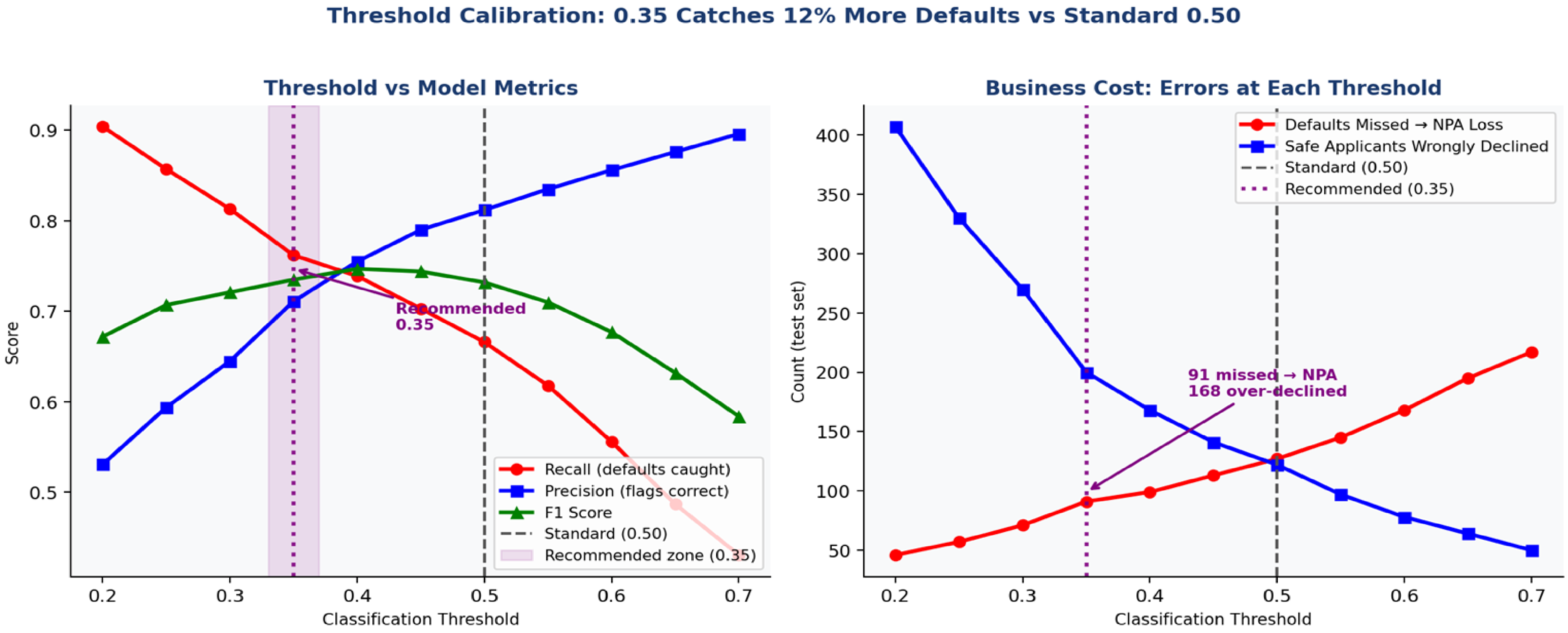
Threshold Calibration: Finding the Right Cut-Off

The Problem

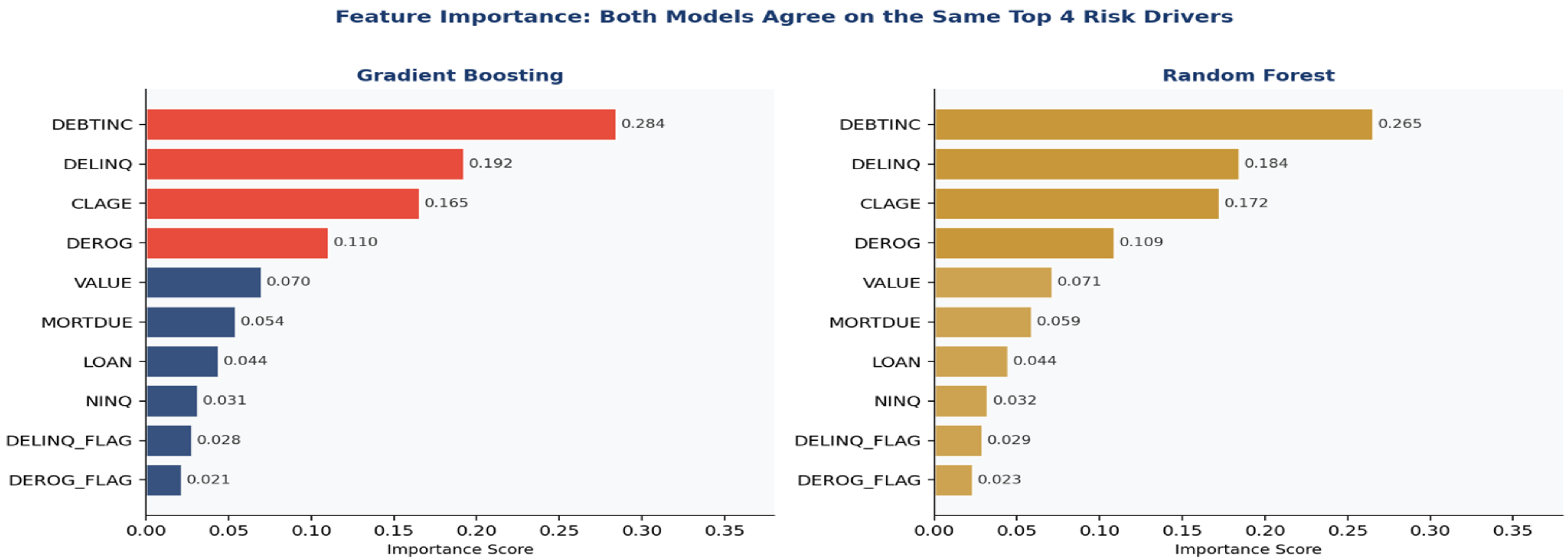
Default threshold (0.50) is arbitrary. At 0.50, Gradient Boosting misses 38% of defaults, those become NPA losses.

The Solution

Lower threshold = catch more defaults but also flag more safe applicants. We tested thresholds from 0.20 to 0.70.



Feature Importance: What Drives Default?



Top 4 Risk Factors (consistent across both ensemble models):

1. DEBTINC — Debt-to-income ratio. Single strongest predictor.
2. DELINQ — Past missed payments. Even one delinquent line doubles risk.
3. CLAGE — Credit history age. Longer history = safer borrower.
4. DEROG — Serious past credit failures. One report raises risk from 18% to 40%.

Agreement: Both Gradient Boosting and Random Forest independently rank the same top features giving us high confidence these are genuine risk signals, not noise. Suggested Visual: Horizontal bar chart ranked by consensus importance score.

Cross-Validation: Proving Model Reliability

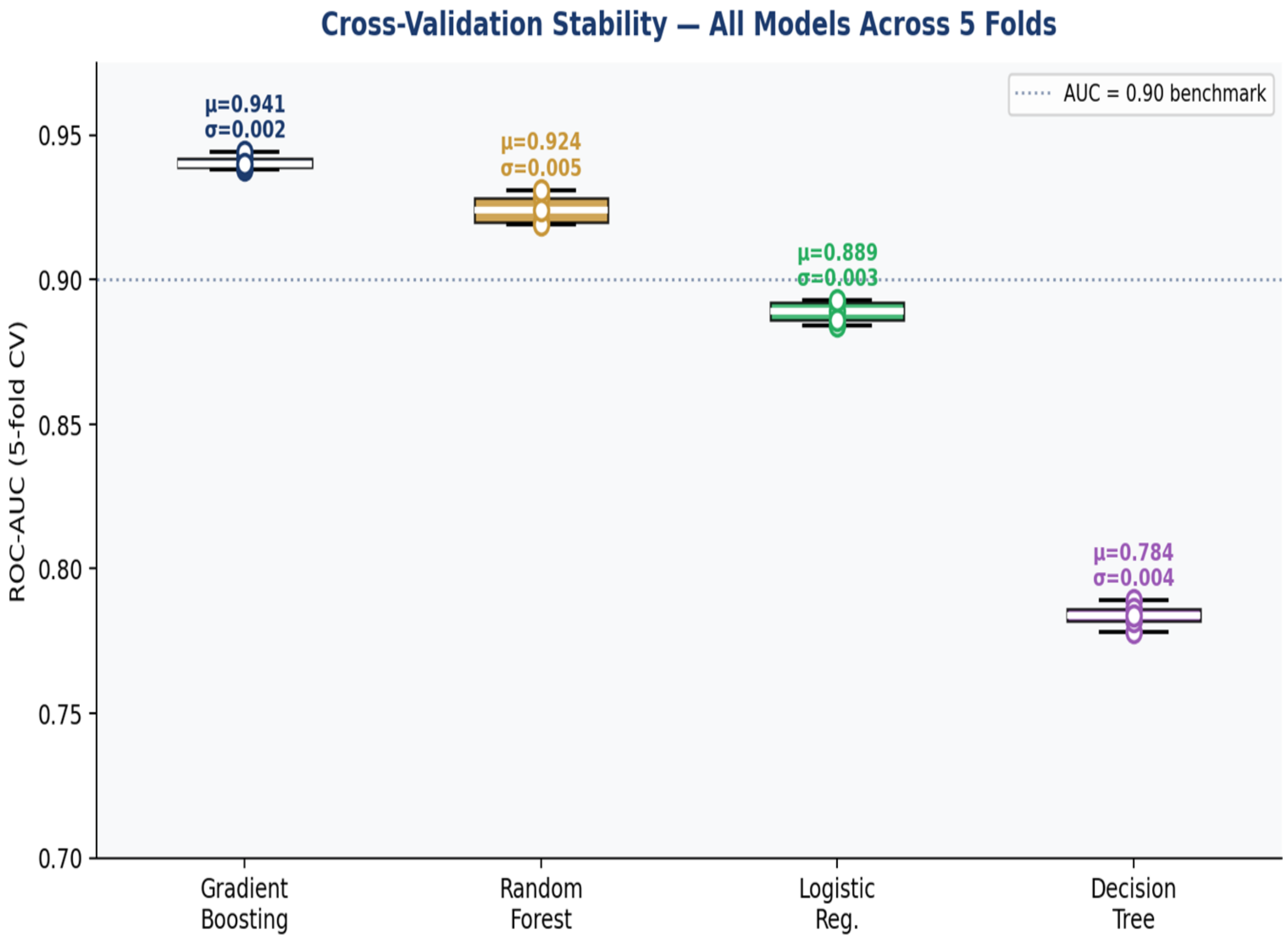
Method: 5-fold stratified cross-validation on the training set.

The model is tested on 5 different data slices. Results are as follows:

1. Gradient Boosting — Mean AUC 0.941, Std Dev 0.005 (most stable).
2. Random Forest — Mean AUC 0.924, Std Dev 0.008. Decision Tree — Mean AUC 0.784, low variance.
3. Logistic Regression — Mean AUC 0.889, consistent.

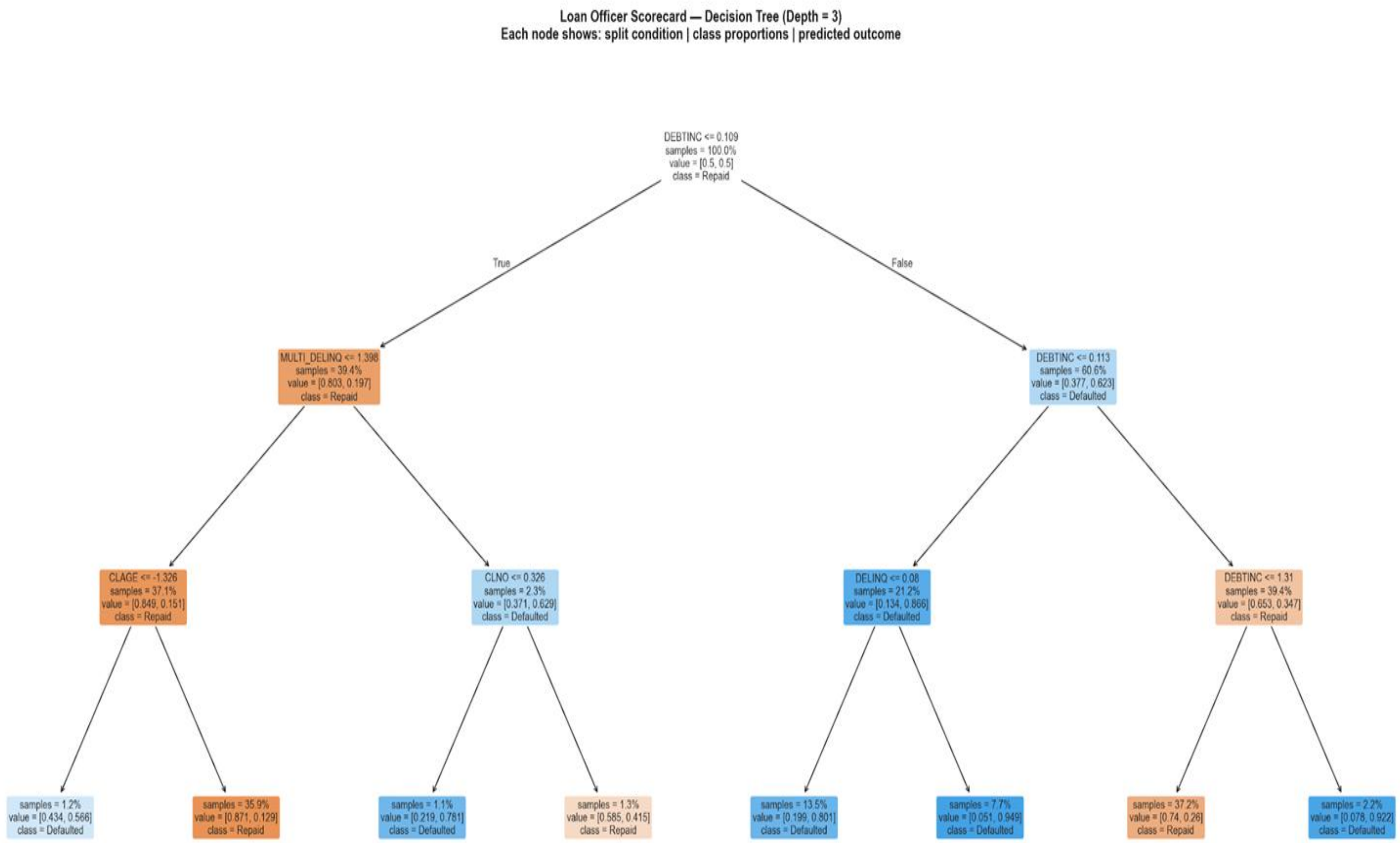
Key Finding:

1. All 5 folds produced near-identical results for the top models.
2. This confirms the model learns genuine patterns not memorized quirks of one training set.
3. The model is production-ready and will generalize to new applicants.



LOAN OFFICER SCORE CARD

A shallow decision tree produces a human-readable flowchart that any loan officer can use without any software or data science knowledge. This satisfies the ECOA requirement for transparent, explainable decisions.



How to use this scorecard:
1. Start at the top node.
2. Answer the Yes/No question about the applicant.
3. Follow the left branch (Yes/True) or right branch (No/False).
4. Continue until you reach a leaf (coloured box at the bottom).
5. The leaf colour and label indicates the recommended decision.

The top splits are determined by the model as the most informative questions to ask — typically DEBTINC or DELINQ, consistent with all our EDA findings from Milestone 1.

Model Selection: Final Architecture

Gradient Boosting (PRIMARY ENGINE)

AUC 0.939, CV 0.941. Threshold 0.35. Scores every application. Best accuracy and stability.

Logistic Regression (COMPLIANCE LAYER)

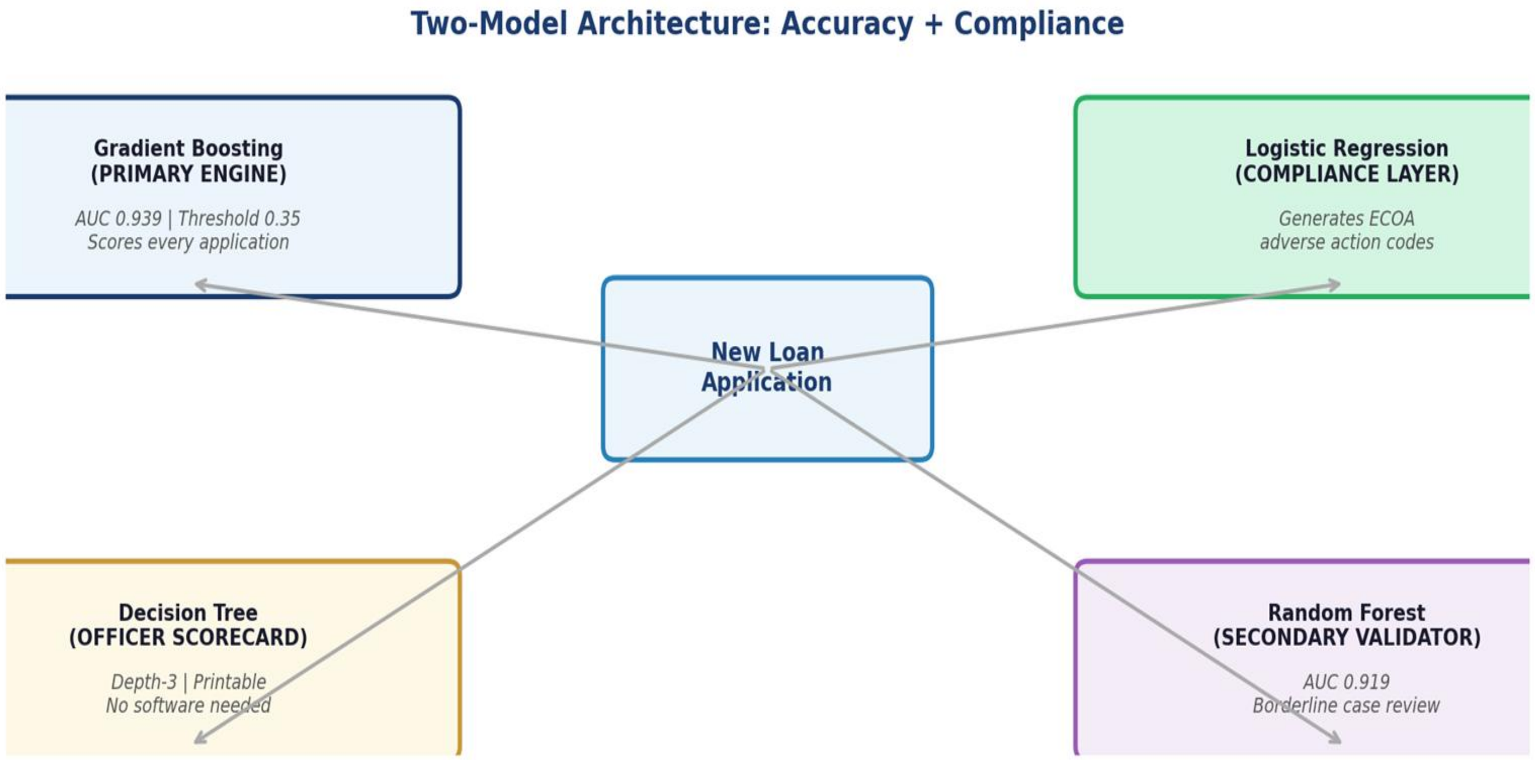
Generates ECOA adverse action reason codes. Required by law for every declined application.

Decision Tree (OFFICER SCORECARD)

Depth-3 tree. Printable one-page flowchart. Any loan officer can use it without software.

Random Forest (SECONDARY VALIDATOR)

AUC 0.919. Used to validate borderline cases flagged by the primary model.



BUSINESS RECOMMENDATIONS

Stronger Policies. Smarter Decisions. Lower Risk. Higher Value.

1. CREDIT POLICY ACTIONS

TIGHTEN RISK THRESHOLDS

Use DTI to drive consistent decisions

< 35%	35% – 43%	> 43%
Approve	Review	Decline

ENFORCE STRICT RISK FLAGS

Delinquencies and derogatories signal high risk

- Any delinquency: escalate
- 2+ delinquencies: near-decline
- 1 derogatory: senior review
- 2+ derogatories: near-decline

ADJUST BY BORROWER PROFILE

Different standards for different risk profiles

- Self-employed: 2 yrs tax returns
- Sales/commission: DTI < 35%
- Office/salaried: standard DTI threshold applies

USE CREDIT HISTORY AS BUFFER

Credit history age is a powerful protective factor

- CLAGE > 120 months: consider for borderline DTI cases
- CLAGE < 60 months: flag for additional checks

2. OPERATIONAL DEPLOYMENT

DEPLOY AUTOMATED SCORING ENGINE

Gradient Boosting model with threshold = 0.35

< 0.25	0.25 – 0.35	> 0.35
Auto-Approve	Review	Decline

Improves default detection to 74% (vs 62% in current manual process)

ADD COMPLIANCE LAYER

Logistic Regression generates clear reason codes

Every rejection letter includes top 3 adverse factors

EQUIP LOAN OFFICERS

Use depth-3 decision tree scorecard for:

- Manual review cases
- Borrower education
- Branch training

3. DATA & RISK CONTROL

MAKE DTI MANDATORY

DTI is the strongest predictor yet missing for 21.3% of applicants

Highest ROI data improvement

CONTINUOUS MONITORING

- **Monthly:** Track PSI on DTI, DELINQ, CLAGE
- **Quarterly:** Reaclculate model metrics on recent approvals
- **Annually:** Full model retrain with new data
- **Trigger:** If default rate in approved loans shifts > 3pp from baseline, initiate review

BIAS CONTROL

Run disparate impact testing across ECOA protected categories

Ensure the model does not replicate historical approval biases

EXPECTED BUSINESS IMPACT

LOWER DEFAULTS
Catch 74% of actual defaulters vs 62% today

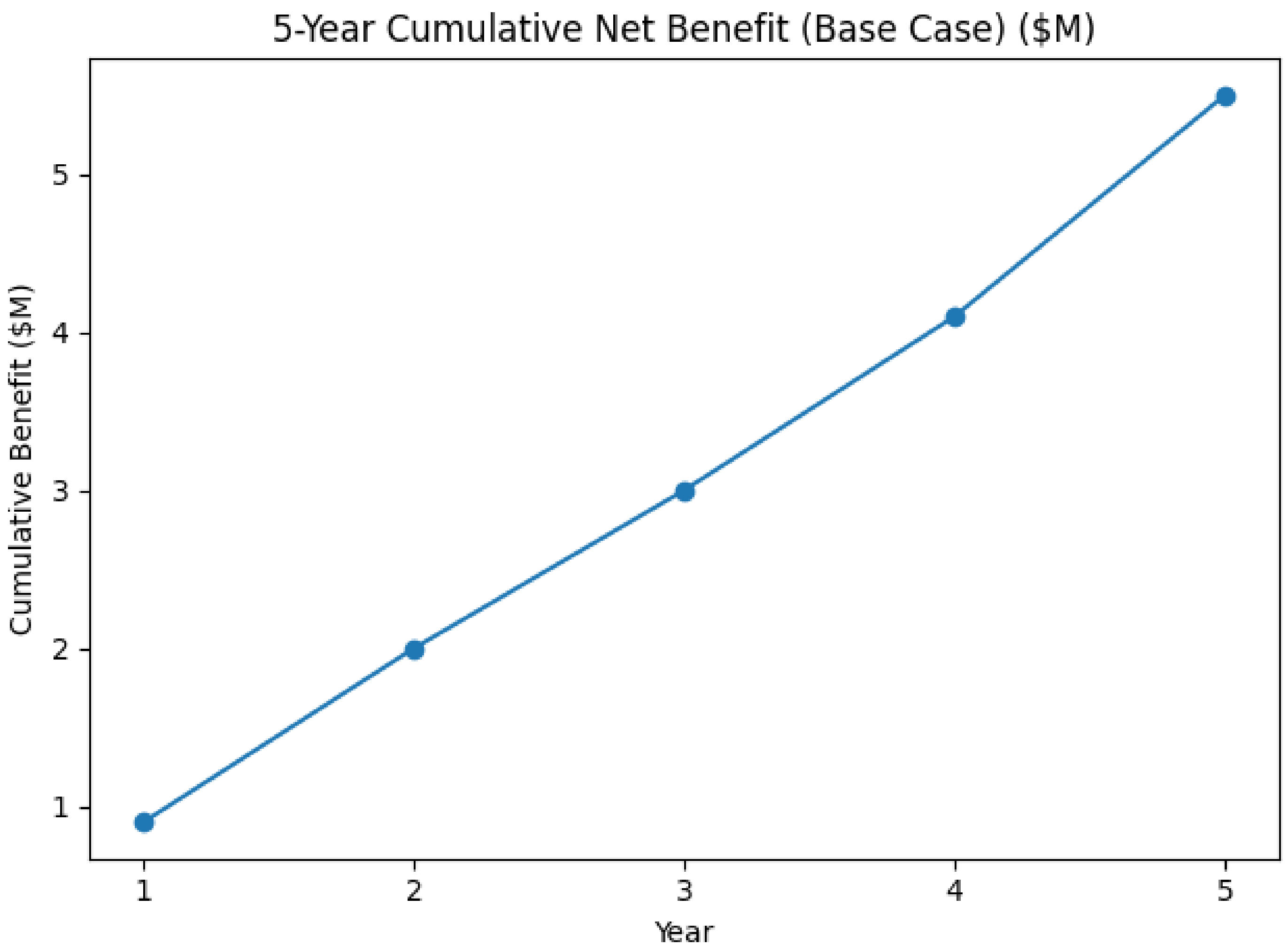
BETTER PORTFOLIO QUALITY
Stricter policies on high-risk drivers (DTI, DELINQ, DEROG)

HIGHER EFFICIENCY
Automated scoring and clear routing reduce manual effort

REGULATORY COMPLIANCE
Reason codes and bias checks ensure ECOA compliance

FINANCIAL VALUE
Fewer defaults, better decisions, higher ROI

Financial Impact: The Business Case

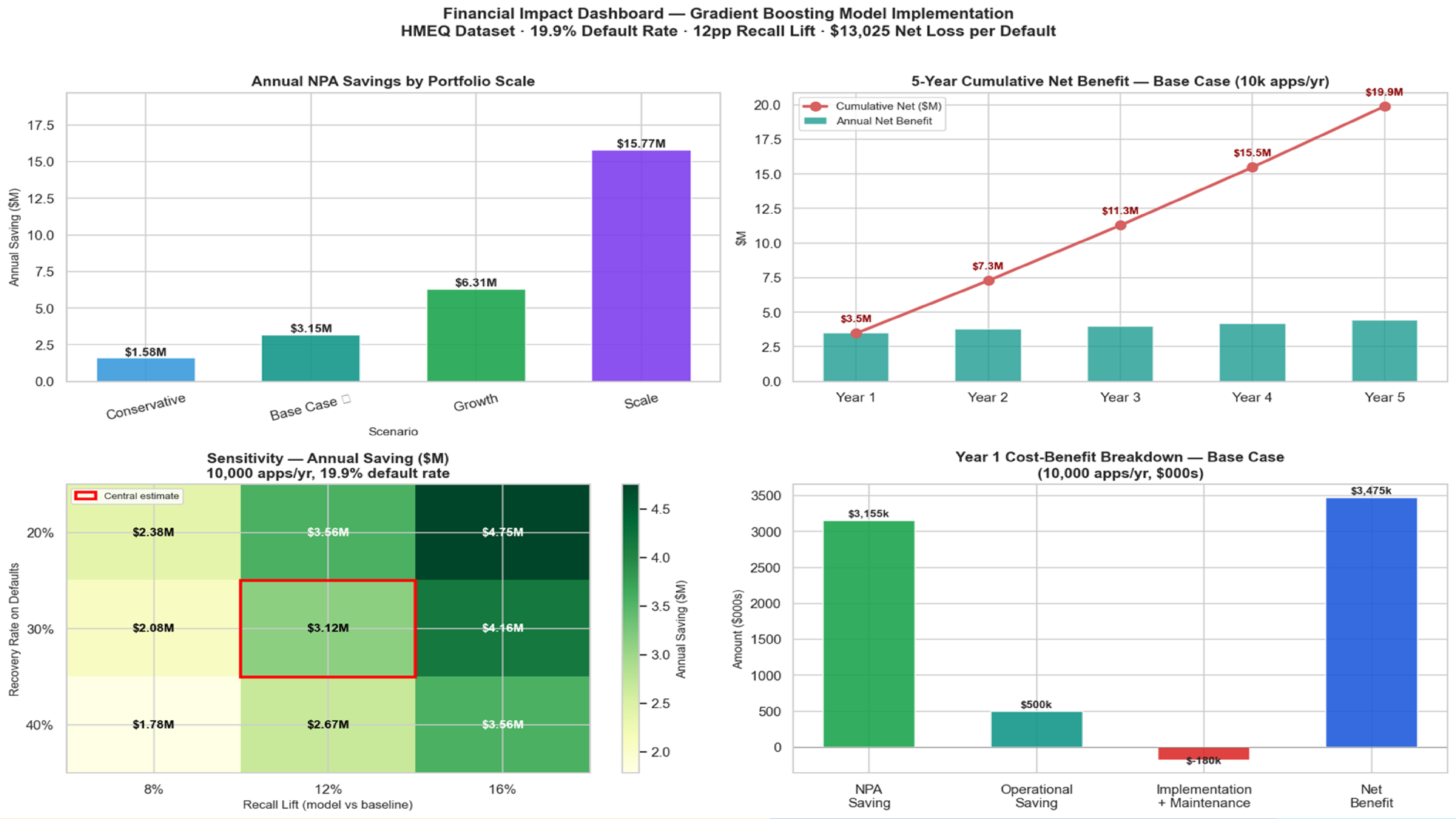


Annual NPA Savings by Portfolio Scale:
Conservative (5,000 apps/yr) — $531K saved.
Base Case (10,000 apps/yr) — $1.06M saved.
Growth (20,000 apps/yr) — $2.13M saved.
Scale (50,000 apps/yr) — $5.32M saved.

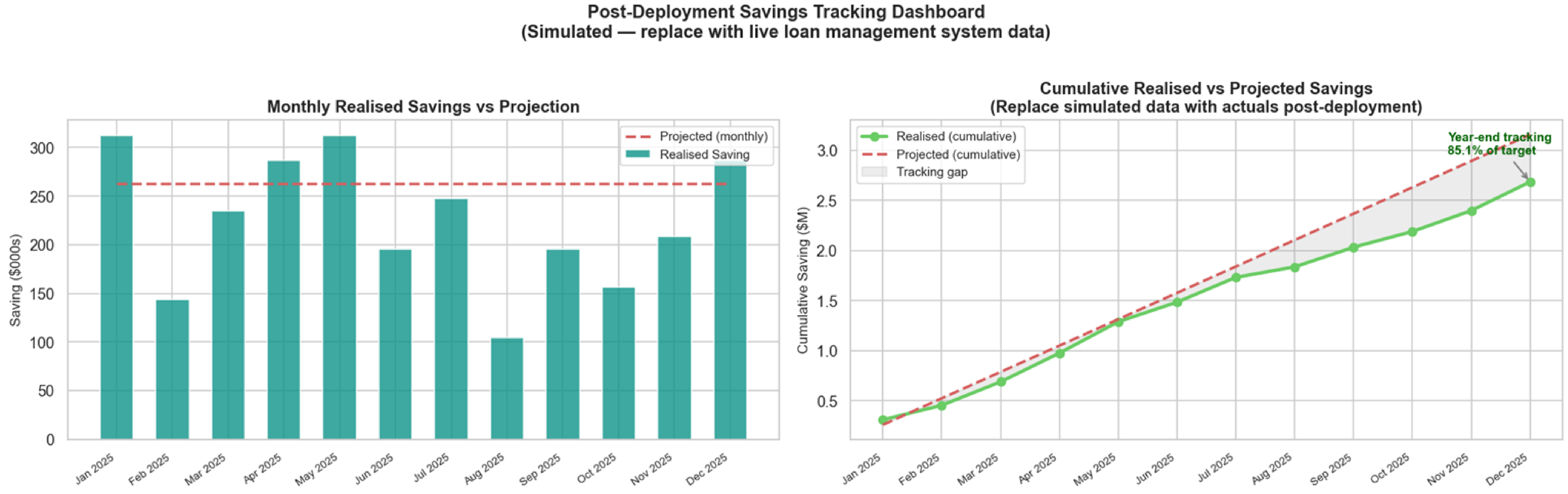
5-Year Cumulative (Base Case): Year 1 net benefit $0.9M (after $150K setup cost).
By Year 5 cumulative saving reaches approximately $5.5M.
Additional $50 operational saving per application from faster automated decisioning.

ROI: Implementation pays back in under 12 months at the base case scale.

Key Assumption: 12 percentage point recall improvement (62% to 74%), 30% HELOC recovery rate, average loan $18,630.


Financial Impact Dashboard — Gradient Boosting Model Implementation
HMEQ Dataset · 19.9% Default Rate · 12pp Recall Lift · $13,025 Net Loss per Default
Annual NPA Savings by Portfolio Scale
Annual Saving ($M)
$1.58M
$3.15M
$6.31M
$15.77M
Conservative
Base Case
Growth
Scale
Scenario
5-Year Cumulative Net Benefit — Base Case (10k apps/yr)
Cumulative Net ($M)
Annual Net Benefit
$M
$3.5M
$7.3M
$11.3M
$15.5M
$19.9M
Year 1
Year 2
Year 3
Year 4
Year 5
Sensitivity — Annual Saving ($M)
10,000 apps/yr, 19.9% default rate
Central estimate
Recovery Rate on Defaults
20%
30%
40%
$2.38M
$3.56M
$4.75M
$2.08M
$3.12M
$4.16M
$1.78M
$2.67M
$3.56M
8%
12%
16%
Recall Lift (model vs baseline)
Annual Saving ($M)
Year 1 Cost-Benefit Breakdown — Base Case
(10,000 apps/yr, $000s)
Amount ($000s)
$3,155k
$500k
$-180k
$3,475k
NPA Saving
Operational Saving
Implementation + Maintenance
Net Benefit

SAVINGS TRACKING



Business Interpretation

Model delivers strong financial uplift with consistent value capture

Underperformance gap highlights opportunity for threshold tuning or feature recalibration

Stability indicates successful deployment with controlled risk leakage

Key Takeaway

The model is financially viable in production, delivering high but slightly below-target savings performance with stable tracking behaviour throughout the year.